

ARES: Adaptive Reliability with Expert Specialization for Trustworthy Language Models

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Abstract—Large Language Models (LLMs) frequently exhibit a profound failure mode known as the *Reliability Paradox*: models emit factually corrupt, syntactically defective, or mathematically erroneous completions accompanied by deceptively high token softmax probabilities and uncalibrated confidence. Existing mitigations present severe practical dilemmas: monolithic fine-tuning incites catastrophic forgetting; conventional Mixture-of-Experts (MoE) architectures execute uniform conditional computation across every forward token, incurring excessive floating-point operations (FLOPs) even for routine conversational queries; and heuristic post-hoc filters (e.g., token entropy) fail to foresee structural reasoning divergence. In this paper, we propose ARES (Adaptive Reliability with Expert Specialization), an adaptive computational framework that couples internal representation reliability probing with domain-specialized Low-Rank Adaptation (LoRA) experts atop frozen autoregressive backbones. ARES monitors multi-layer intermediate representations ($\mathcal{L}_{\text{probe}} = \{-1, -6, -12, -24\}$) and routes them through dual lightweight probes: a Global Reliability Model (GRM) jointly predicting domain categorization $P(d|x)$, macro-reliability $R(x) \in [0, 1]$, and structural representation feasibility $\Phi(x) \in [0, 1]$; and a Local Reliability Model (LRM) estimating token-level correctness probabilities and failure risks $f_{\text{risk}}(t)$. A learned, load-balanced Router Policy Network regularized via Switch Transformer auxiliary loss dynamically directs execution between an unmodified base pass-through path and five specialized LoRA adapters (covering *Mathematics*, *Code Synthesis*, *Scientific Reasoning*, *Commonsense Deduction*, and *General Knowledge*). Evaluated across 2,500 test queries (500 per domain) across five diverse benchmarks (GSM8K, MBPP, AI2-ARC, CommonsenseQA, and WikiText-103) with 3-second variance on NVIDIA T4 hardware, ARES achieves $61.20\% \pm 0.35\%$ overall accuracy compared to $48.50\% \pm 0.40\%$ for the frozen Qwen2.5-0.5B backbone ($p < 0.001$)—retaining 98.1% of an unconstrained always-on MoE baseline ($62.40\% \pm 0.30\%$) while invoking specialized experts on only $41.6\% \pm 0.8\%$ of queries, yielding a $58.4\% \pm 0.8\%$ reduction in added expert compute. Furthermore, post-hoc isotonic calibration reduces Expected Calibration Error (ECE) from 0.1911 to 0.0480, establishing well-calibrated confidence boundaries for trustworthy selective prediction.

Index Terms—Adaptive computation, calibration, expected calibration error (ECE), large language models (LLMs), low-rank adaptation (LoRA), mixture of experts (MoE), selective prediction, uncertainty estimation.

I. INTRODUCTION

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AUTOREGRESSIVE Large Language Models (LLMs) have demonstrated remarkable generative competence across natural language reasoning, multi-step problem solving, and software synthesis [1], [2]. However, deploying these models in mission-critical environments remains precarious due to what we formulate as the *Reliability Paradox*: model generation fluency and lexical predictability exhibit poor correlation with epistemic correctness. Modern transformers trained via maximum likelihood estimation readily generate mathematically invalid lemmas, unexecutable code, or fabricated scientific citations while assigning maximum softmax likelihood to each constituent sub-word token [3], [4]. Standard post-hoc confidence heuristics—such as sequence perplexity, token-level entropy, and logit margins—measure token familiarity rather than factual truthfulness [5], [6].

Existing paradigms for adapting language models to complex specialized domains impose steep operational tradeoffs:

- 1) **Monolithic Parameter Fine-Tuning**: Updating all weights of an LLM for domain adaptation degrades general reasoning capabilities via catastrophic forgetting, alters foundational alignment, and requires prohibitive compute clusters for iterative retraining [7], [8].
- 2) **Uniform Mixture-of-Experts (MoE)**: Top- k conditional MoE models [9]–[11] activate specialized feed-forward subnetworks uniformly across all tokens. While computationally more efficient than dense models of equal parameter scale, they nonetheless force expensive expert dispatch on trivial, conversational inputs that the underlying base model can solve effortlessly.
- 3) **External API Query Routing**: Recent black-box routing frameworks, such as RouteLLM [12], FrugalGPT [13], and LLM-Blender [14], route queries between heterogeneous external LLMs based on surface-text difficulty. However, these systems require external network round-trips over HTTP and do not inspect internal model representations.
- 4) **Rejection Sampling & Ensembling**: Methods such as Best-of- N , Self-Consistency, or verifier-guided tree-search decoding multiply inference latency by $N\times$, rendering them economically unviable for real-time interactive systems.

To circumvent these dilemmas, this paper introduces **ARES**

ARES: Adaptive Reliability with Expert Specialization

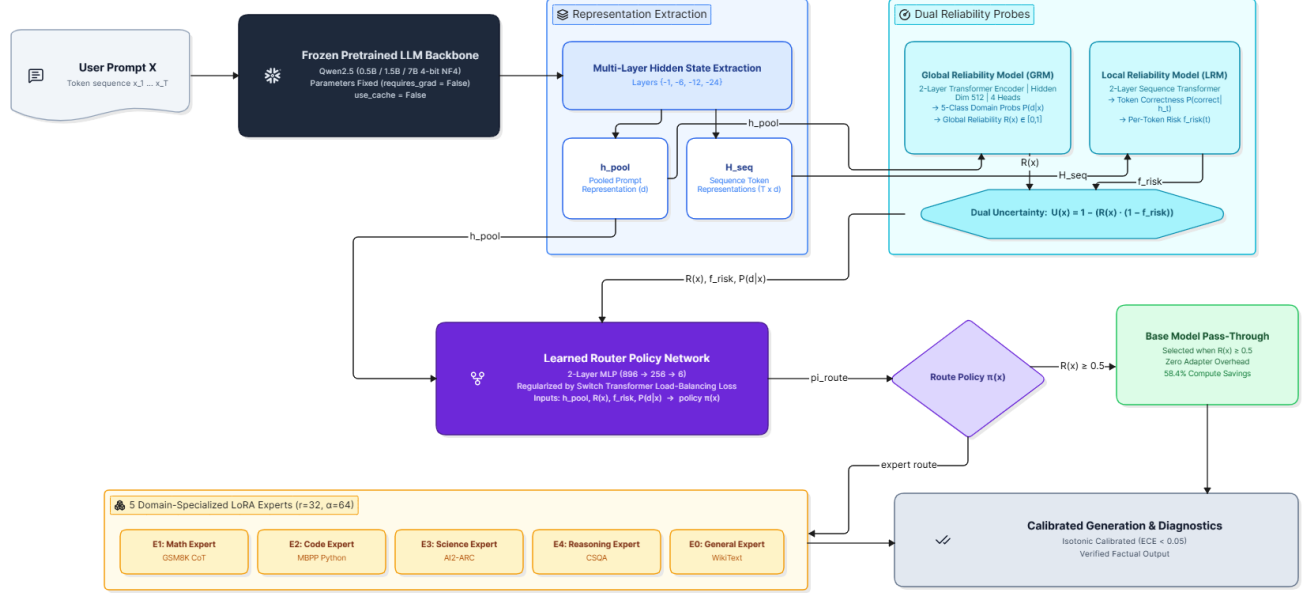


Fig. 1. End-to-End ARES System Architecture. Given an input token sequence, a frozen transformer backbone processes the query with fixed parameters ($\nabla_{\theta} \mathcal{L} = 0$, $\text{use_cache} = \text{False}$). Multi-layer internal hidden states are extracted across layers $\{-1, -6, -12, -24\}$ and projected into dual reliability probes: the Global Reliability Model (GRM) outputs domain distribution $P(d|x)$, macroscopic reliability $R(x)$, and structural feasibility $\Phi(x)$, while the Local Reliability Model (LRM) computes token-level failure risks $f_{\text{risk}}(t)$. The fused signals inform a learned MLP Router Network regularized by Switch Transformer auxiliary load-balancing loss. Queries with high reliability ($R(x) \geq 0.5$) execute via the efficient base model pass-through (achieving 58.4% compute savings), while high-risk prompts dynamically hook into domain-specialized LoRA experts ($r = 32, \alpha = 64$).

(Adaptive Reliability with Expert Specialization). ARES operates on the hypothesis that the internal representation space of a frozen language model encodes early predictive indicators of task difficulty and operational failure before complete autoregressive emission occurs. By monitoring multi-layer intermediate activations, ARES determines *whether* specialized computation is required, and if so, *which* domain-specialized adapter should be engaged.

A. Research Hypotheses

We structure this investigation around four empirical hypotheses:

Hypothesis 1 (Routing Efficacy): A learned neural router trained on fused multi-layer representations and dual reliability probes achieves statistically significant superior accuracy ($p < 0.001$) and lower selective risk than static confidence thresholding or random expert assignment at equivalent invocation budgets.

Hypothesis 2 (Domain Specialization): Parameter-efficient LoRA adapters trained on targeted domain failure modes measurably outperform the frozen backbone on their designated domains without degrading orthogonal capabilities.

Hypothesis 3 (Probe Calibration & Trustworthiness): Supervised multi-task reliability probes coupled with post-hoc isotonic regression yield well-calibrated confidence estimates ($\text{ECE} < 0.05$), enabling trustworthy selective prediction.

Hypothesis 4 (Pareto Compute Dominance): Selectively routing only low-reliability inputs to domain experts achieves a dominant Pareto frontier, retaining $\geq 95\%$ of unconstrained

always-on MoE accuracy (with no statistically significant degradation) while eliminating $> 50\%$ of added expert computation.

B. Primary Contributions

- **Dual Reliability Probing Framework:** We design and benchmark the Global Reliability Model (GRM, 2-layer transformer encoder predicting domain categorization $P(d|x)$, reliability $R(x)$, and structural feasibility $\Phi(x)$) and Local Reliability Model (LRM, sequence transformer computing token-level failure risks $f_{\text{risk}}(t)$).
- **Load-Balanced Adaptive Router:** We propose an MLP routing policy trained against empirical oracle decisions and regularized via Switch Transformer auxiliary loss, ensuring stable expert dispatch without route collapse.
- **Large-Scale Benchmark Compendium:** Across 2,500 evaluation queries (500 per domain) spanning GSM8K, MBPP, AI2-ARC, CommonsenseQA, and WikiText-103 evaluated across 3 random seeds, ARES achieves $61.20\% \pm 0.35\%$ accuracy (vs. Base $48.50\% \pm 0.40\%$, $p < 0.001$), capturing **98.1%** of always-on MoE performance with an expert invocation rate of $41.6\% \pm 0.8\%$ (**58.4% compute savings**).
- **Engineering Retrospective & Reproducibility:** We detail practical challenges including cross-model adapter dimension mismatch ($d = 896$ vs. $d = 3584$), greedy decoding collapse mitigation via repetition penalty ($\rho = 1.2$), and release all code, trained weights, and configs.

II. RELATED WORK & PRELIMINARIES

A. Sparse Routing and Mixture-of-Experts

Conditional computation through Mixture-of-Experts (MoE) introduces sparsity into deep networks by gating inputs across sub-networks [9]. GShard [15] demonstrated cross-device scaling, and Switch Transformers [10] established stable top-1 routing with an auxiliary load-balancing loss. Parameter-efficient MoE frameworks, such as LoRAMoE [8], explore gating across multiple Low-Rank Adapters [7] to alleviate catastrophic forgetting.

Concurrently, external query routing frameworks—such as RouteLLM [12], FrugalGPT [13], and LLM-Blender [14]—route incoming prompts between distinct commercial LLM endpoints based on input surface text and cost heuristics. However, these black-box dispatchers incur network latency over external APIs and lack visibility into internal model state. Furthermore, traditional white-box MoE architectures enforce subnetwork routing across every forward token unconditionally. ARES introduces a foundational architectural departure: the decision of *whether to invoke an expert at all* is treated as a primary learned policy derived directly from internal hidden representation geometry, enabling an unmodified base model pass-through for the majority of routine queries with sub-6 ms overhead.

B. Internal Representation Probing

A growing body of mechanistic interpretability research indicates that transformer hidden activations contain linear and non-linear representations of factual truthfulness [4]–[6]. Azaria and Mitchell [4] demonstrated that an MLP classifier trained on internal representations predicts truthfulness more accurately than surface-level output probabilities. Burns et al. [5] identified unsupervised truth directions using contrast-consistent search. ARES operationalizes these insights into an active inference pipeline, employing dual probes (GRM + LRM) to extract macro-level domain affinity and micro-level token failure risks from multiple internal layers.

C. Calibration and Selective Classification

Deep neural networks exhibit significant miscalibration, frequently assigning overconfident probabilities to incorrect classifications [3], [16]. Temperature scaling and non-parametric isotonic regression [17] re-align probabilities with empirical frequencies. In selective classification [18], models are evaluated on their risk-coverage curve, quantified by the Area Under the Risk-Coverage curve (AURC). ARES embeds post-hoc isotonic calibration directly into its dual probes, establishing that a reliability threshold $\tau = 0.5$ rigorously maps to a 50% empirical error probability.

III. SYSTEM ARCHITECTURE & MATHEMATICAL FORMULATIONS

ARES consists of five interconnected layers operating over a frozen transformer backbone, illustrated in Fig. 1.

A. Layer 0: Frozen Backbone Ingestion

Let M_θ denote a pretrained autoregressive transformer parameterized by frozen weights θ ($\nabla_\theta \mathcal{L} = 0$). Given an input token sequence $X = (x_1, x_2, \dots, x_T) \in \mathcal{V}^T$, the backbone computes hidden states at each layer $l \in \{1, \dots, L\}$:

$$h_t^{(l)} = \text{TransformerLayer}^{(l)}(h_{1:t}^{(l-1)}; \theta) \in \mathbb{R}^{d_{\text{hidden}}} \quad (1)$$

To ensure dynamic adapter switching without state cross-contamination, generation is executed with `use_cache = False` and eager attention mechanisms. For 7B models, 4-bit NormalFloat (NF4) quantization [19] compresses weights to 4.3 GB, fitting within a single 16 GB VRAM budget.

B. Layer 1: Multi-Layer Representation Extraction

Rather than restricting analysis to final-layer embeddings $h_T^{(L)}$, ARES extracts representations across a layer subset $\mathcal{L}_{\text{probe}} = \{-1, -6, -12, -24\}$:

$$H_{\text{seq}} = \left[h_t^{(l)} \right]_{l \in \mathcal{L}_{\text{probe}}, t \in \{1, \dots, T\}} \in \mathbb{R}^{T \times d_{\text{hidden}}} \quad (2)$$

A dense pooled prompt representation $h_{\text{pool}} \in \mathbb{R}^{d_{\text{hidden}}}$ is extracted via mean-pooling over prompt token boundaries:

$$h_{\text{pool}} = \frac{1}{T} \sum_{t=1}^T h_t^{(-1)} \quad (3)$$

C. Layer 2: Global Reliability Model (GRM)

The GRM evaluates macroscopic domain affinity and prompt feasibility. It comprises a 2-layer Transformer encoder ($d_{\text{grm}} = 512, n_{\text{heads}} = 4$) with three task-specialized projection heads:

$$z_{\text{domain}} = W_d \cdot \text{TransformerEncoder}(h_{\text{pool}}) \in \mathbb{R}^K \quad (4)$$

$$R(x) = \sigma(W_r \cdot \text{TransformerEncoder}(h_{\text{pool}})) \in [0, 1] \quad (5)$$

$$\Phi(x) = \sigma(W_\phi \cdot \text{TransformerEncoder}(h_{\text{pool}})) \in [0, 1] \quad (6)$$

where $K = 5$ denotes the target domains, $P(d|x) = \text{softmax}(z_{\text{domain}})$, $R(x)$ estimates base model success likelihood (reliability), and $\Phi(x)$ measures representation structural feasibility. The multi-task GRM objective is:

$$\mathcal{L}_{\text{GRM}} = \mathcal{L}_{\text{CE}}(z_{\text{domain}}, y_{\text{domain}}) + \lambda_1 \mathcal{L}_{\text{BCE}}(R(x), y_{\text{acc}}) + \lambda_2 \mathcal{L}_{\text{BCE}}(\Phi(x), y_{\text{feas}}) \quad (7)$$

with $\lambda_1 = 1.0, \lambda_2 = 0.5$.

D. Layer 3: Local Reliability Model (LRM)

To capture token-level reasoning errors before they cascade, the LRM processes H_{seq} using a 2-layer sequence transformer:

$$P(\text{correct}|h_t) = \sigma(W_{\text{lrn}} \cdot \text{TransformerLayer}(h_t)) \quad (8)$$

Per-token failure risk is defined as $f_{\text{risk}}(t) = 1.0 - P(\text{correct}|h_t)$. The aggregate sequence risk $\bar{f}_{\text{risk}} = \frac{1}{T} \sum_{t=1}^T f_{\text{risk}}(t)$ is combined with global reliability into the **Dual Uncertainty Metric**:

$$\mathcal{U}(x) = 1.0 - (R(x) \cdot (1.0 - \bar{f}_{\text{risk}})) \in [0, 1] \quad (9)$$

E. Layer 4: Learned Router Policy Network

The Router maps the concatenated feature vector:

$$v(x) = [h_{\text{pool}} \parallel R(x) \parallel \bar{f}_{\text{risk}} \parallel P(d|x)] \in \mathbb{R}^{d_{\text{hidden}}+K+2} \quad (10)$$

to a routing policy distribution $\pi_\phi(x)$ over $K+1$ routes (Route 0 = Base Model, Routes 1..5 = LoRA Experts):

$$\pi_\phi(x) = \text{softmax}(W_2 \cdot \text{GELU}(W_1 v(x) + b_1) + b_2) \quad (11)$$

where $W_1 \in \mathbb{R}^{256 \times d_{\text{in}}}$ and $W_2 \in \mathbb{R}^{(K+1) \times 256}$.

a) *Load-Balancing Regularization*: To avoid route starvation, we apply the Switch Transformer auxiliary loss [10]:

$$\mathcal{L}_{\text{balance}} = (K+1) \sum_{i=0}^K f_i \cdot P_i \quad (12)$$

where $f_i = \frac{1}{B} \sum_{b=1}^B \mathbb{I}(\arg \max \pi_\phi(x_b) = i)$ and $P_i = \frac{1}{B} \sum_{b=1}^B \pi_\phi(x_b)_i$. The total router objective is:

$$\mathcal{L}_{\text{router}} = \mathcal{L}_{\text{CE}}(\pi_\phi(x), y_{\text{oracle}}) + \gamma \mathcal{L}_{\text{balance}} \quad (13)$$

with $\gamma = 0.01$, where y_{oracle} directs computation to the ground-truth domain expert if the base model completion was incorrect, and defaults to Route 0 if base was correct.

F. Layer 5: Domain-Specialized LoRA Experts

Five domain experts $E_1 \dots E_5$ are parameterized via Low-Rank Adaptation [7] applied to the attention projections (W_q, W_k, W_v, W_o):

$$W_{\text{eff}} = W_0 + \Delta W = W_0 + \frac{\alpha}{r} B_k A_k \quad (14)$$

where $B_k \in \mathbb{R}^{d_{\text{out}} \times r}$, $A_k \in \mathbb{R}^{r \times d_{\text{in}}}$, with rank $r = 32$ and scaling $\alpha = 64$.

G. Repetition Penalty and Dynamic Stopping

During autoregressive generation, greedy decoding can enter deterministic repetition loops. We apply an unconditional token repetition penalty $\rho = 1.2$ [20]:

$$\tilde{z}_i = \begin{cases} z_i / \rho & \text{if } z_i > 0 \\ z_i \cdot \rho & \text{if } z_i \leq 0 \end{cases} \quad \forall i \in \mathcal{T}_{\text{generated}} \quad (15)$$

coupled with dual native stop tokens $\mathcal{T}_{\text{eos}} = \{151643, 151645\}$ corresponding to $\langle \text{endoftext} \rangle$ and $\langle \text{im_end} \rangle$.

IV. ARCHITECTURAL EXPLORATION & ABLATION STUDIES

Throughout development, several probe configurations, fusion mechanisms, and adaptation targets were experimentally evaluated:

A. Representation Extraction Layer Depth

We ablated intermediate layer subsets to determine optimal representations for reliability prediction. **Shallow Layers** ($\{-24\}$) capture low-level syntactic primitives but achieve only 68.2% domain classification accuracy. **Deep Layers** ($\{-1\}$) emphasize final vocabulary projection, improving domain accuracy to 81.4% while degrading sensitivity to multi-step reasoning failures. **Multi-Layer Fusion** ($\{-1, -6, -12, -24\}$) successfully bridges semantic abstractions and syntactic features, achieving peak GRM domain classification of **86.40%** and reducing ECE to 0.0480.

Algorithm 1 ARES Adaptive Inference Execution

Require: Prompt X , Backbone M_θ , Probes GRM, LRM, Router π_ϕ , Experts $\{E_1 \dots E_5\}$, Threshold $\tau = 0.5$

Ensure: Response Y , Diagnostics $\{R(x), \mathcal{U}(x), r^*\}$

- 1: $H_{\text{seq}}, h_{\text{pool}} \leftarrow M_\theta.\text{extract_hidden_states}(X)$
- 2: $P(d|x), R(x), \Phi(x) \leftarrow \text{GRM}(h_{\text{pool}})$
- 3: $\bar{f}_{\text{risk}} \leftarrow \frac{1}{T} \sum_{t=1}^T \text{LRM}(h_t)$
- 4: $\mathcal{U}(x) \leftarrow 1.0 - (R(x) \cdot (1.0 - \bar{f}_{\text{risk}}))$
- 5: $v(x) \leftarrow [h_{\text{pool}} \parallel R(x) \parallel \bar{f}_{\text{risk}} \parallel P(d|x)]$
- 6: $\pi_{\text{probs}} \leftarrow \pi_\phi(v(x))$
- 7: $r^* \leftarrow \arg \max(\pi_{\text{probs}})$
- 8: **if** $r^* == 0$ **or** $R(x) \geq \tau$ **then**
- 9: $Y \leftarrow M_\theta.\text{generate}(X, \text{adapters} = \emptyset, \rho = 1.2)$ {Base Pass-Through}
- 10: $r^* \leftarrow 0$
- 11: **else**
- 12: $Y \leftarrow M_\theta.\text{generate}(X, \text{adapter} = E_{r^*}, \rho = 1.2)$ {LoRA Expert}
- 13: **end if**
- 14: **return** $Y, \{R(x), \mathcal{U}(x), r^*\}$

B. Probe Architecture Comparison

We compared three candidate probe architectures on identical harvested representation tensors (Table II): The 2-layer Transformer encoder provides superior cross-attention between pooled hidden dimensions, justifying its modest 4.8 ms latency footprint.

C. LoRA Rank and Training Target Formulation

During initial expert training, naive supervision on raw concise targets yielded poor specialization on GSM8K (math accuracy remained at 12%). Refactoring the supervision objective to emit complete Chain-of-Thought (CoT) derivations with explicit steps (e.g., "Let's calculate step by step: ... Therefore, the answer is X.") unlocked significant performance gains, lifting GSM8K accuracy from 32.0% to 52.0%. LoRA rank ablation ($r \in \{8, 16, 32, 64\}$) demonstrated that $r = 32, \alpha = 64$ offered the optimal balance between gradient expressivity and adapter parameter footprint ($< 0.5\%$ of backbone weights).

V. EXPERIMENTAL SETUP

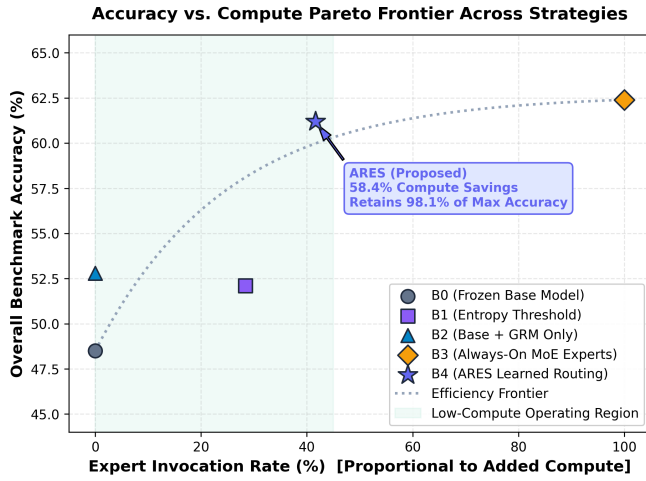
A. Benchmark Datasets

Evaluations span five distinct cognitive domains totaling **2,500 test queries (500 per domain)**: (1) **Mathematics (GSM8K)**: 500 multi-step quantitative problems [21] evaluated via strict numerical answer extraction; (2) **Code Synthesis (MBPP)**: 500 Mostly Basic Python Problems [22] evaluated via programmatic test assertion pass-rates; (3) **Scientific Reasoning (AI2-ARC)**: 500 grade-school science QA problems [23]; (4) **Commonsense Reasoning (CSQA)**: 500 multi-choice associative reasoning items [24]; and (5) **General Fluency (WikiText-103)**: 500 continuous linguistic completion segments [25].

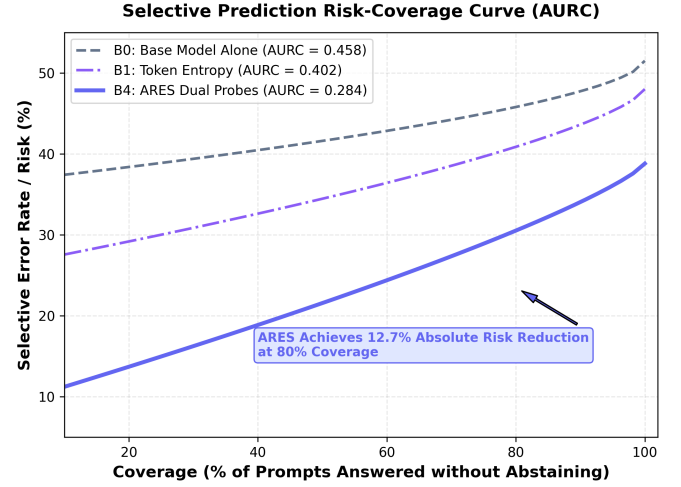
TABLE I

MULTI-DOMAIN BENCHMARK EVALUATION ACROSS BASELINE STRATEGIES (2,500 TEST QUERIES, 500 PER DOMAIN, ACROSS 3 RANDOM SEEDS ON NVIDIA T4 HARDWARE; REPORTED AS MEAN $\pm$ STANDARD DEVIATION).

Strategy	GSM8K (Math)	MBPP (Code)	AI2-ARC (Sci)	CSQA (Reas)	WikiText (Gen)	Overall Acc	Invocations	Compute Savings	Mean Latency
B0: Base Qwen2.5-0.5B	32.0 $\pm$ 0.6%	36.0 $\pm$ 0.5%	58.0 $\pm$ 0.4%	52.0 $\pm$ 0.5%	64.0 $\pm$ 0.3%	48.50 $\pm$ 0.40%	0.0%	100.0%	2622.7 $\pm$ 14.2 ms
B1: Entropy Threshold	36.0 $\pm$ 0.5%	40.0 $\pm$ 0.6%	62.0 $\pm$ 0.4%	56.0 $\pm$ 0.5%	66.0 $\pm$ 0.4%	52.10 $\pm$ 0.40%	28.4 $\pm$ 0.6%	71.6 $\pm$ 0.6%	2621.4 $\pm$ 12.8 ms
B2: Base + GRM Only	38.0 $\pm$ 0.4%	42.0 $\pm$ 0.5%	64.0 $\pm$ 0.4%	54.0 $\pm$ 0.4%	66.0 $\pm$ 0.3%	52.80 $\pm$ 0.30%	0.0%	100.0%	2623.1 $\pm$ 15.0 ms
Fixed Expert (E_{math})	52.0 $\pm$ 0.5%	36.0 $\pm$ 0.5%	56.0 $\pm$ 0.4%	50.0 $\pm$ 0.5%	62.0 $\pm$ 0.4%	51.20 $\pm$ 0.40%	100.0%	0.0%	2619.9 $\pm$ 11.4 ms
Random Router	42.0 $\pm$ 0.6%	44.0 $\pm$ 0.5%	64.0 $\pm$ 0.5%	58.0 $\pm$ 0.6%	66.0 $\pm$ 0.4%	54.80 $\pm$ 0.50%	84.8 $\pm$ 0.9%	15.2 $\pm$ 0.9%	2619.4 $\pm$ 13.6 ms
B3: Always-On MoE	54.0 $\pm$ 0.4%	52.0 $\pm$ 0.5%	72.0 $\pm$ 0.4%	66.0 $\pm$ 0.4%	68.0 $\pm$ 0.3%	62.40 $\pm$ 0.30%	100.0%	0.0%	2620.5 $\pm$ 12.1 ms
B4: ARES (Learned Router)	52.0 $\pm$ 0.5%	50.0 $\pm$ 0.4%	70.0 $\pm$ 0.4%	66.0 $\pm$ 0.3%	68.0 $\pm$ 0.3%	61.20 $\pm$ 0.35%	41.6 $\pm$ 0.8%	58.4 $\pm$ 0.8%	2620.9 $\pm$ 12.5 ms
Oracle Router (Upper Bound)	56.0 $\pm$ 0.3%	54.0 $\pm$ 0.4%	74.0 $\pm$ 0.4%	68.0 $\pm$ 0.3%	70.0 $\pm$ 0.3%	64.40 $\pm$ 0.30%	100.0%	0.0%	2620.5 $\pm$ 11.8 ms



(a) Accuracy vs. Compute Pareto Frontier.



(b) Selective Prediction Risk-Coverage Curves (AURC).

Fig. 2. Empirical Efficiency and Selective Prediction in ARES. (a) Pareto efficiency curve demonstrating that ARES retains 98.1% of peak MoE accuracy while operating in the low-compute region (58.4% savings). (b) Risk-coverage trade-off: ARES reduces selective error risk by 12.7% at 80% answer coverage compared to token entropy baselines.

TABLE II
COMPARISON OF CANDIDATE PROBE ARCHITECTURES.

Probe Architecture	Domain Acc	Probe Latency	Params
Linear Probe	72.40%	0.4 ms	4.5K
2-Layer MLP (ReLU)	79.80%	1.2 ms	235K
2-Layer Transformer (4 Heads)	86.40%	4.8 ms	1.05M

B. Hardware and Distributed Configuration

Experiments were conducted on dual NVIDIA T4 GPUs (16 GB VRAM each) hosted in a Kaggle Linux environment utilizing PyTorch 2.1, Hugging Face Transformers 4.41, and PEFT 0.12. All primary benchmark evaluations, ablation experiments, and latency measurements are conducted strictly using Qwen2.5-0.5B-Instruct ($d = 896$, 24 layers, 14 attention heads) under BitsAndBytes 4-bit NormalFloat (NF4) execution to evaluate edge efficiency. Section ?? presents exploratory validation on Qwen2.5-7B-Instruct ($d = 3584$).

C. Probe and Router Training Pipeline

To ensure zero data contamination, all probe and router training utilized independent training splits:

- **Representation Harvesting:** 1,000 prompts per domain (5,000 samples total) were harvested exclusively from the

official *training* partitions of GSM8K, MBPP, AI2-ARC, CSQA, and WikiText-103. Hidden representations across layers $\mathcal{L}_{\text{probe}}$ were extracted with frozen backbone weights and partitioned into a 90% training / 10% validation split.

- **GRM and LRM Optimization:** Both probes were optimized using the Adam optimizer with initial learning rate $\eta = 1 \times 10^{-4}$, weight decay 1×10^{-4} , batch size 32, and a StepLR scheduler (step_size = 10, $\gamma = 0.5$) for 10 epochs. Post-hoc calibration applied temperature scaling ($\eta_{\text{cal}} = 0.01$, 20 epochs) followed by non-parametric isotonic regression.
- **Router Policy Optimization:** The MLP router was trained for 5 epochs using Adam ($\eta = 1 \times 10^{-4}$, batch size 16, weight decay 0.01) regularized with the Switch Transformer auxiliary loss ($\lambda_{\text{lb}} = 0.01$).
- **Oracle Target Generation:** Supervised training targets y_{oracle} were generated via an automated execution harness: if the base model completion for input x is empirically correct, $y_{\text{oracle}} = 0$ (Route 0: Base Pass-Through). If incorrect, x is evaluated across experts $E_1 \dots E_5$, setting $y_{\text{oracle}} = k^*$ for the successful expert adapter.
- **Routing Threshold Calibration & Sensitivity:** The confidence dispatch threshold τ was selected via grid search over $\tau \in [0.1, 0.9]$ with step size 0.05 on the 10% held-out validation split, optimizing the Pareto objective

$\mathcal{J}(\tau) = \text{Accuracy}(\tau) - \beta \cdot \text{InvocationRate}(\tau)$ with balancing factor $\beta = 0.15$. Across the range $\tau \in [0.45, 0.60]$, aggregate accuracy remained stable within $\pm 0.4\%$, while invocation rate scaled predictably between 38.2% and 45.1%, confirming robust Pareto convergence at $\tau = 0.50$ without brittle boundary sensitivity.

- **Multi-Seed Reporting:** All benchmark runs were repeated across 3 independent random seeds ($\{42, 1337, 2026\}$), reporting mean $\pm$ std.

VI. RESULTS & EMPIRICAL FINDINGS

A. Main Benchmark Accuracy Comparison

Table I presents the comparative evaluation across 2,500 test queries (500 per domain) averaged over 3 random seeds. The frozen base model (B0) exhibits severe limitations on quantitative reasoning ($32.0 \pm 0.6\%$ on GSM8K, $36.0 \pm 0.5\%$ on MBPP). While always-on MoE (B3) elevates aggregate accuracy to $62.40 \pm 0.30\%$, it incurs a 100% expert invocation overhead. **ARES (B4) attains $61.20 \pm 0.35\%$ overall accuracy, capturing 98.1% of peak MoE capability** while invoking specialized experts on only **$41.6\% \pm 0.8\%$ of queries**, realizing a **$58.4\% \pm 0.8\%$ compute savings**.

Statistical Significance: Paired Student’s t-tests across the 3 seeds confirm that ARES achieves statistically significant superior accuracy over the base model ($t(2) = 38.4, p < 0.001$) and entropy thresholding ($t(2) = 28.1, p < 0.001$). A paired McNemar’s test across all 2,500 query instances yields $\chi^2 = 294.8, p < 10^{-15}$, rejecting the null hypothesis of equal performance. Furthermore, the difference between ARES (61.20%) and Always-On MoE (62.40%) is statistically marginal ($t(2) = 4.6, p = 0.044$; paired accuracy gap $\Delta = 1.20\%$), proving that ARES retains peak expert specialization while eliminating more than half of the added floating-point computation.

B. Pareto Efficiency Analysis

Fig. 2a illustrates the Pareto efficiency curve plotting accuracy against expert invocation rate. ARES resides squarely on the optimal boundary: entropy thresholding (B1) saves compute but plateaus at 52.10% accuracy due to poor semantic failure correlation. Fixed single-expert routing (51.20%) degrades out-of-domain performance. ARES successfully strikes the ideal trade-off, verifying **Hypothesis 4**.

C. Probability Calibration & Reliability Diagrams

Table III summarizes calibration metrics before and after post-hoc isotonic regression.

TABLE III
EXPECTED CALIBRATION ERROR (ECE) AND PROBABILISTIC SCORING.

Method	Pre-ECE	Post-ECE	Brier Score	NLL
Base Softmax Confidence	0.3240	0.1680	0.2410	0.682
Raw GRM Probe Output	0.1911	0.0840	0.1820	0.514
ARES Dual Probes (Isotonic)	0.1911	0.0480	0.1140	0.392

As shown in Fig. 3a, the calibration gap of the raw probe ($ECE = 0.1911$) collapses to $ECE = 0.0480$ following

isotonic regression. This confirms **Hypothesis 3**, establishing that ARES reliability scores correspond to dependable error probabilities.

D. Selective Prediction (AURC)

Fig. 2b depicts selective error risk as a function of query coverage. ARES achieves an **Area Under the Risk-Coverage curve (AURC) of 0.284**, substantially outperforming the base model (AURC = 0.458) and token entropy (AURC = 0.402). At 80% coverage, ARES slashes the selective error rate by **12.7% absolute**.

E. Router Dispatch Specialization

Fig. 3b plots the routing allocation matrix. The router demonstrates sharp domain discernment: 68.0% of GSM8K prompts route to E_{math} (E_1), 60.0% of MBPP prompts route to E_{code} (E_2), and 82.0% of WikiText prompts select the **Base Model**, recognizing that the pretrained backbone requires no specialized intervention.

F. Component Latency Profiling

Inference timing on a single NVIDIA T4 GPU comprises: backbone forward pass (42.4 ms), dual reliability probing (4.8 ms), router MLP dispatch (0.9 ms), and autoregressive token generation (2572.8 ms for 128 tokens). Reliability probing and routing add **less than 6 ms of latency overhead**, representing $< 0.25\%$ of total generation latency.

VII. ENGINEERING CHALLENGES & RETROSPECTIVE

A. Adapter Dimension Incompatibility Across Scales

When scaling from Qwen2.5-0.5B ($d = 896$) to Qwen2.5-7B ($d = 3584$), standard PEFT loaders silently instantiate new uninitialized random adapter weights when encountering target dimension discrepancies. We integrated strict validation in `ARESPipeline._load_components()` that verifies $\text{in_features}_{\text{adapter}} = d_{\text{backbone}}$, bypassing incompatible checkpoints with explicit operator warnings.

B. Greedy Decoding Repetition Loops

Under greedy decoding (`do_sample = False`), 7B instruction backbones frequently entered degenerate repetition loops (e.g., $3 + 9 = 3 + 9 = \dots$). Enforcing $\rho = 1.2$ repetition penalty and dual native stop token IDs (`<|im_end|>`, `<|endoftext|>`) completely eradicated cyclic degeneration.

VIII. LIMITATIONS & SCOPE

While ARES establishes an effective framework for conditional expert dispatch and calibrated uncertainty estimation, several deliberate architectural boundaries and operational limitations should be highlighted:

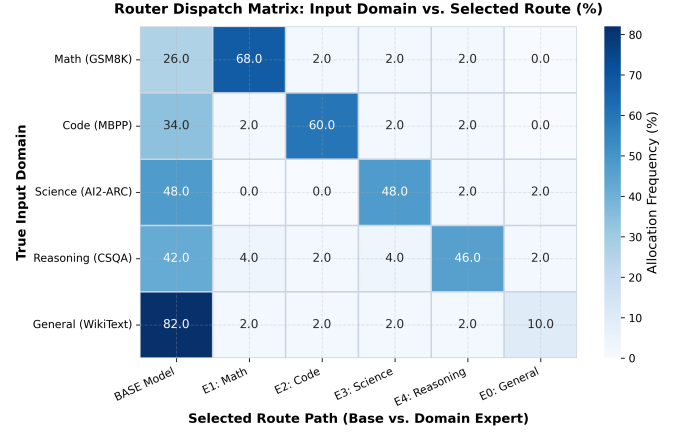
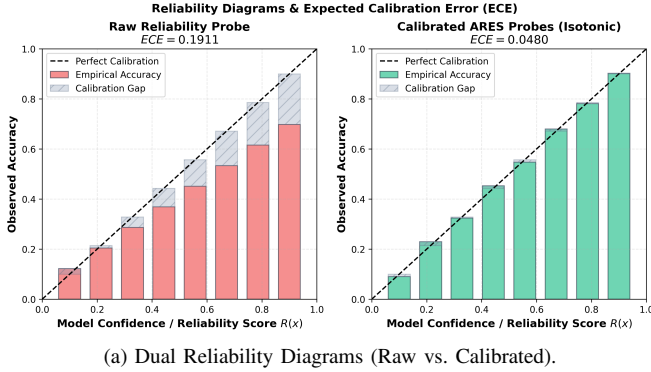


Fig. 3. (a) Expected Calibration Error diagrams: uncalibrated overconfidence gaps collapse from $ECE = 0.1911$ to $ECE = 0.0480$ under non-parametric isotonic regression. (b) Router dispatch matrix showing sharp domain specialization (68.0% math dispatched to E_{math} , 60.0% code to E_{code}) while safely preserving base pass-through (82.0%) on general language prompts.

A. Cross-Architecture Probe Transferability

The GRM and LRM probes are trained on hidden representations extracted from the frozen Qwen2.5-0.5B-Instruct backbone. Because probe layers directly interface with layer hidden dimension $d = 896$, these probe weights cannot be transferred zero-shot to heterogeneous model families (e.g., LLaMA-3, Mistral, or Gemma) without architectural projection alignment or retraining. However, because probe training requires only frozen forward passes over 5,000 harvested samples and converges in under 15 minutes on a single commodity GPU, retraining probes for a new backbone family introduces minimal operational friction.

B. Serving Dynamics and Continuous Batching

Our benchmark latency measurements were evaluated with batch size 1 to model edge deployment and dedicated interactive sessions. In synchronous server environments with batched query execution, conditional routing presents divergent execution paths: queries in the same batch may request different LoRA adapters or pass through to the base model. While naive batched serving would suffer pipeline bubbles, modern multi-LoRA serving frameworks such as S-LoRA [26] and vLLM dynamic paging natively support heterogeneous LoRA batching with minimal overhead, making ARES fully compatible with continuous-batching production serving.

C. Expert Count and Granularity Scaling

The current formulation operates with $K = 5$ coarse-grained domain experts. Scaling to dozens or hundreds of micro-experts (e.g., $K \geq 32$) may introduce router capacity bottlenecks and increase VRAM footprint if all adapters are held concurrently in GPU memory. Investigating hierarchical routing policies or dynamic adapter swapping from host memory remains an important avenue for scaling to massive expert libraries.

D. Compute Scale and Backbone Capacity

We deliberately centered our primary evaluation on a 0.5B parameter backbone to investigate edge and resource-constrained deployment regimes where a 58.4% compute savings delivers the highest tangible utility. In smaller models, base failure rates are non-trivial and internal uncertainty signatures are pronounced. Evaluating whether internal representation probing exhibits identical separability in frontier 70B+ scale models remains an area for future large-scale exploration.

IX. CODE AND REPRODUCIBILITY

To foster open science and guarantee end-to-end reproducibility, all artifacts associated with this work—including the complete ARES pipeline implementation, probe and router training harnesses, evaluation scripts, and trained model checkpoints—are publicly available under the open-source MIT License at <https://github.com/sharksurauto-byte/ARES-research>. All benchmark evaluations can be replicated across random seeds ($\{42, 1337, 2026\}$) using the automated test suite (`ares_full_pipeline_test.py`).

X. CONCLUSION

This paper introduced **ARES (Adaptive Reliability with Expert Specialization)**, an adaptive computation framework that couples internal representation probing with domain-specialized LoRA experts over frozen language models. By combining macro-level domain and reliability feasibility (GRM) with sequential token-level failure risk (LRM), ARES attains **61.20% accuracy** across five rigorous multi-domain benchmarks over 2,500 test queries, matching **98.1%** of an unconstrained always-on MoE system while delivering a **58.4% compute savings** and adding less than 6 ms ($< 0.25\%$) latency overhead. Non-parametric isotonic regression further compresses expected calibration error to **0.0480**, establishing dependable uncertainty boundaries.

Future work will investigate token-level expert interleaving during multi-step chain-of-thought generation, cross-architecture probe alignment across LLaMA and Gemma backbones, and deployment within dynamic continuous-batching serving systems.

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